

A close-up, artistic photograph of a dental casting machine. The image shows a rotating metal part, likely a mold or a component of the machine, with a polished, reflective surface. The lighting is dramatic, with strong highlights and deep shadows, emphasizing the metallic texture and the precision of the machinery. The background is blurred, focusing attention on the mechanical details.

Dental casting alloys

Marwa Bahaa

Lecturer of dental biomaterials

Classification of Dental Casting Alloys

1. According to nobility:

High-noble metal alloys (at least 40% gold and at least 60 % noble metals).

Noble metal alloys (more than 25 % noble metal).

Base metal alloys (less than 25 % noble metal).

Classification of Dental Casting Alloys

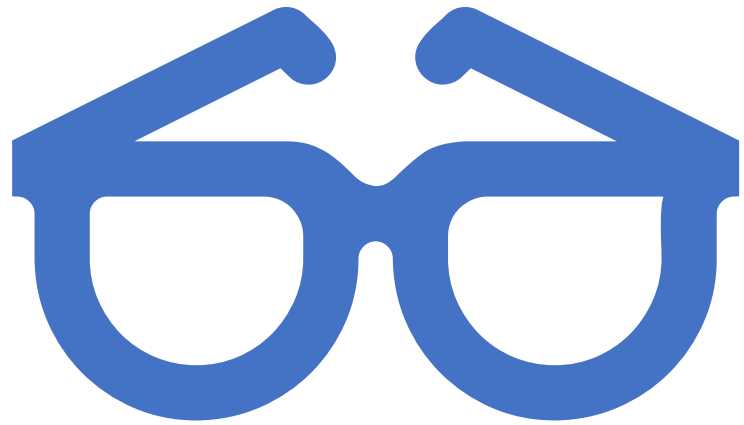
2. According to mechanical properties:

- Type I (soft) restorations subjected to slight stress as inlays.
- Type II (medium) restorations subjected to moderate stresses as onlays.
- Type III (hard) restorations subjected to high stresses as crown.
- Type IV (extra hard) restorations subjected to very high stresses as long span bridges and RPD framework.

Classification of Dental Casting Alloys

3. According to use:

- Alloys for all metal restorations.
- Alloys for ceramo-metallic restorations.
- Alloys for RPD framework.



**High Noble
Alloys**

Noble Elements

	Gold	Platinum	Palladium	Silver	Iridium
Melting point	1063°C	1777°C	1555°C	960°C	2440°C
Specific gravity	29.3	21.37	11.4	10.4	22.4

Noble Elements

	Gold	Platinum	Palladium	Silver	Iridium
Color	yellow	white	white	white	
Properties	Soft, ductile	Tough, ductile	Ductile	Ductile	
Strength , hardness	low	High	High	Higher than gold	

Noble Elements

Gold	Platinum	Palladium	Silver	Iridium
Resist tarnish & corrosion.	- Increase strength & hardness. - Resist tarnish & corrosion.	- Increase strength & hardness. - Whiten the alloy. - Resist tarnish & corrosion. - Prevent silver corrosion.	Increase strength & hardness.	Grain refiner (0.005%).

Base Elements

	Copper	Zinc	Indium	Tin
Melting point	1083°C	419°C	156°C	232°C
Role	• Increase strength & hardness. • Important in heat treatment.	• Scavenger (chelating agent). • Improve castability and fluidity of the alloy.	• Replacement of zinc. • Produce oxide layer for bonding with porcelain (in PFM).	• Produce oxide layer for bonding with porcelain (in PFM).

Classifications of Gold Alloys

1. According to Gold content:

a) *Carat specification:*

- It is the part of pure gold in 24 parts of alloy.
- 24 carat = pure gold.
- 21 carat = 21 parts gold +3 parts other metal.



Classifications of Gold Alloys

1. According to Gold content:

b) *Fineness:*

- It is the part of pure gold in 1000 parts of alloy.
- 750 = 750 parts gold + 250 part other metals.

Classifications of Gold Alloys

2. According to mechanical properties:

- Type I (soft) restorations subjected to slight stress as inlays.
- Type II (medium) restorations subjected to moderate stresses as onlays.
- Type III (hard) restorations subjected to high stresses as crown.
- Type IV (extra hard) restorations subjected to very high stresses as long span bridges and RPD framework.

Classifications of Gold Alloys



3. According to alloy description:

a) *Description by color:*

- **Yellow gold**: the palladium contents is not high enough to conceal the yellow color of gold.
- **White gold**: the palladium content is increased above 5 % so it whitening the alloy . The alloy composition is mainly gold.

Classifications of Gold Alloys

3. According to alloy description:

b).Description based on low gold content (economy gold):

- The gold content is decreased to the range 42%to 58% in order to decrease the cost of the gold alloy.
- The gold is replaced by silver.
- The silver : palladium ratio is important (3:1) to prevent tarnish of silver.(why??)
- The silver: copper ratio is important to prevent tarnish of silver.
- Its mechanical properties are comparable to type III and IV gold alloys.

Noble alloys

Porcelain-fused
to-metal (PFM)



Ceramic
crowns



Gold alloy
crowns



Base metal
alloy crowns



Zirconia
crowns



Noble Alloys

They are less expensive than gold alloys.

They have low density which represents difficulties during casting.

Noble Alloys

1. Silver- Palladium alloys:

- The silver content is more than palladium content.
- They contain no gold or very little amount of gold (10-15%).
- Their mechanical properties are comparable to type III and IV gold alloys.

Noble Alloys

2. Palladium-Silver alloys:

- The palladium content is more than silver content (60% and 30% respectively).



Noble Alloys

Problems with their use:

- Lower density → greater centrifugal force for casting.
- Lower ductility → lower burnishability.
- Tendency to gas absorption during melting & casting → porosity they need specialized equipment for melting & casting.

Gold- copper system

Gold-Copper system

	Gold	Copper
Crystal structure	FCC	FCC

No chemical reaction between Gold & Copper

Gold-Copper system

	Disordered Solid Solution	Ordered Solid Solution
Rate of cooling	Rapid cooling (quenching)	Slow cooling
Strength Hardness Proportional limit	Low	High
Ductility	High	Low
Superlattice		AuCu AuCu ₃

Gold-Copper system

	AuCu ₃	AuCu
Space lattice	Face centered cubic (like Au & Cu)	Face centered tetragonal.
Gold content	39.6-64.1%	64.1-88.4%
Cooling	Slow cooling under 396°C	Slow cooling under 424°C
Strength	Less increased as the space lattice of the AuCu ₃ superlattice like the parent alloy	More increased due to difference in the space lattice between parent alloy and new superlattice (AuCu) why??

Gold-Copper system

Softening heat treatment	Hardening heat treatment	
Heat the alloy at 700°C for 10 minutes then quenched	Heat the alloy at 700°C for 10 minutes then quench	Disordered structure
	- Heat the alloy at 450°C for 2 minutes - Slow cooling until 250°C quench	Ordered structure (AuCu or AuCu₃) according to gold content.

Gold-Copper System

For dental
applications:

- AuCu₃ superlattice is not used in dental applications due to: why??
 - High copper content → low tarnish & corrosion resistance
 - Same spacelattice of parent alloy → little improvement of mechanical properties.

Gold-Copper System

For dental applications:

- Gold alloys type III and IV **can** be subjected to hardening heat treatment due to presence of sufficient copper content. **Why??**
- Gold alloys type I and II **cannot** be subjected to hardening heat treatment due to low copper content.

Why??

Base metal alloys



Base Metal Alloys

They contain **no** noble metals.

So they are **cheaper** than high noble and noble alloys.

Their properties are comparable to gold alloys type III and IV.

Base Metal Alloys

Types:

- Cobalt Chromium alloy.
- Cobalt Chromium nickel alloy.
- Nickel Chromium alloy.
- Titanium and Titanium alloy.

Base Metal Alloys

Composition and role of major elements(= 90 %)

1. Cobalt(35-65 %):

- Increase strength ,hardness and modulus of elasticity of the alloy.

2. Chromium (28-30%):

- It forms **passive layer** → increase tarnish & corrosion resistance.
- If less than 12% → no passive layer formation.
- If more than 30% → produce brittle alloy.

Why??

Base Metal Alloys

3. Nickel (0-30 %):

- Increase strength, hardness and modulus of elasticity but less than cobalt (used interchangeably with cobalt).
- Increase ductility of the alloy.
- Allergic to some patients.

Base Metal Alloys

4. *Titanium:*

- Light weight.
- Low modulus of elasticity (compared to cobalt).
- High strength.
- Form passive layer (if removed, it will return in nanoseconds).
- It is available as commercially pure titanium (CPT) and titanium alloy.



Base Metal Alloys

Composition and role of minor elements (=10%)

1. *Molybdenum*:

- Grain refiner → increase strength.

2. *Silicon and Manganese*:

- Deoxidizers.
- Improve fluidity and castability of the alloy.

Base Metal Alloys

3. *Carbon (0.4 %):*

- Forms carbides → precipitate at grain boundaries → increase strength
- If more than 2% → forms brittle and hard alloy

4. *Aluminum:*

- Forms nickel aluminate → increase strength

Base Metal Alloys

5. Beryllium :

- Lowers melting temperature of the alloy by 100 °C
- Improve fluidity and castability.
- Its vapor is carcinogenic (to technician) → not used anymore

Dental Casting Alloys (biocompatibility)

Gold alloys	Base metal alloys
• Biocompatible due to nobility of gold and other noble metals.	• Biocompatible due to presence of passive layer. • Nickel is allergic to some patients. • Beryllium vapor is carcinogenic.

Dental Casting Alloys (physical properties)

	Gold alloys	Base metal alloys
Tarnish & corrosion resistance	Resistance due to nobility of the metals	Resistance due to presence of passive layer
Color	Yellow color(gold & copper %) White color (palladium , platinum and silver %)	Lustrous silvery white
Melting range	800-1050 °C	Co-Cr =1500°C Ni-Cr = 1300°C Ti=1700°C
Density	15-18	Co-Cr and Ni-Cr = 7 Ti=4.5
Coefficient of thermal expansion	14.8	Co-Cr and Ni-CR = 14 Ti=9
Casting shrinkage	1.6%	Co_Cr and Ni-Cr =2.3% Ti= 3.5%

Dental Casting Alloys (mechanical properties)

	Gold alloys	Base metal alloys
Modulus of elasticity	Low	High
Yield strength	Low	High
Hardness	Low	High
% elongation(ductility)	high	low

Dental Casting Alloys (casting)

	Gold alloys	Base metal alloys
Investment	• Gypsum bonded investment. • Phosphate bonded investment (mixed with water).	• Carbon free phosphate bonded investment (mixed with silica). • Silicate bonded investment.
Casting ring	Metallic ring	Rubber ring
Crucible	Carbon or ceramic	• Ceramic • Carbon not used to avoid carbon contamination which will produce brittle material why??

Dental Casting Alloys (casting)

	Gold alloys	Base metal alloys
Melting	Gas-air torch	• Co-Cr and Ni-Cr → oxyacetylene • Ti → electric melting
Casting machine	Air pressure casting	• Co-Cr and Ni-Cr → centrifugal casting • Ti → centrifugal with air pressure under vacuum (due to low density)
Finishing & polishing	Acid pickling with warm HCl then polishing with rubber cups and polishing paste.	• Sandblasting • Electrolytic polishing (acid pickling should not be used to avoid removing of passive layer). why??

Thank You

